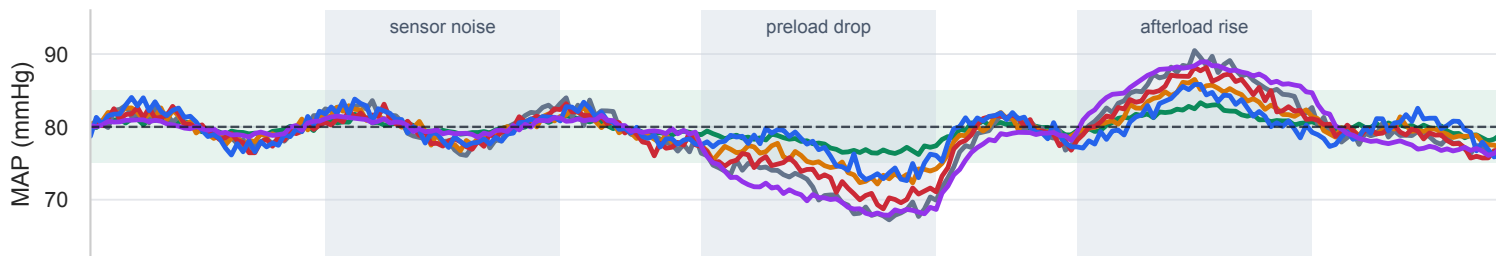
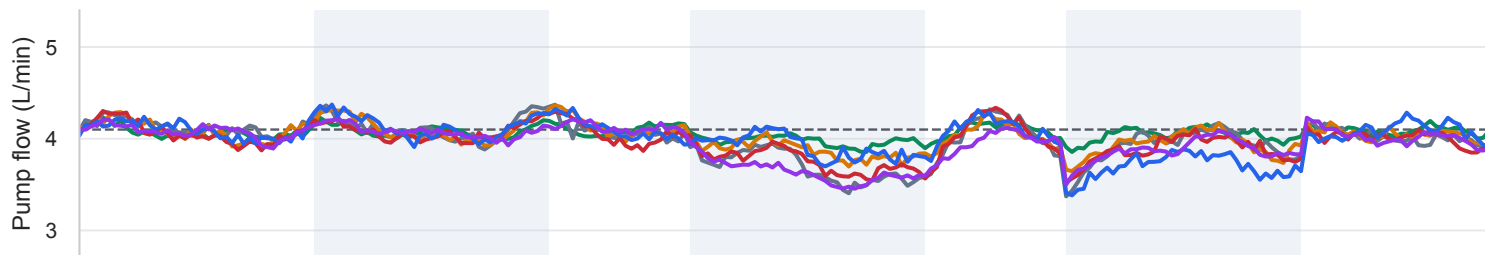


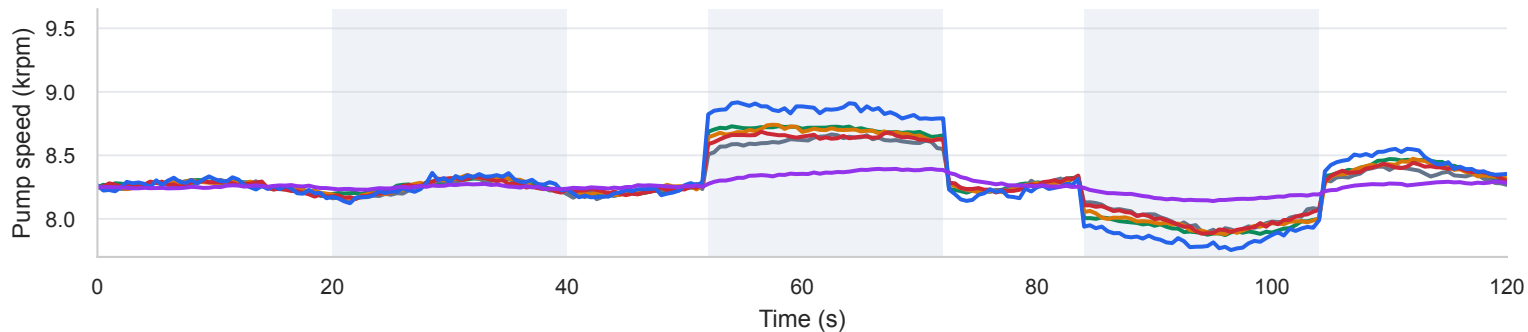
Target MAP versus actual MAP under composite disturbances



Pump flow under disturbances



Pump speed command



Legend:

- Proposed CNN-Transformer (Green line)
- Transformer-only Actor-Critic (Orange line)
- Reward ablation baseline (Blue line)
- MLP Actor-Critic (Dark Blue line)
- CNN-only Actor-Critic (Red line)
- Fixed-speed baseline (Purple line)

Dashed lines denote target MAP or nominal flow; gray windows indicate disturbance intervals.